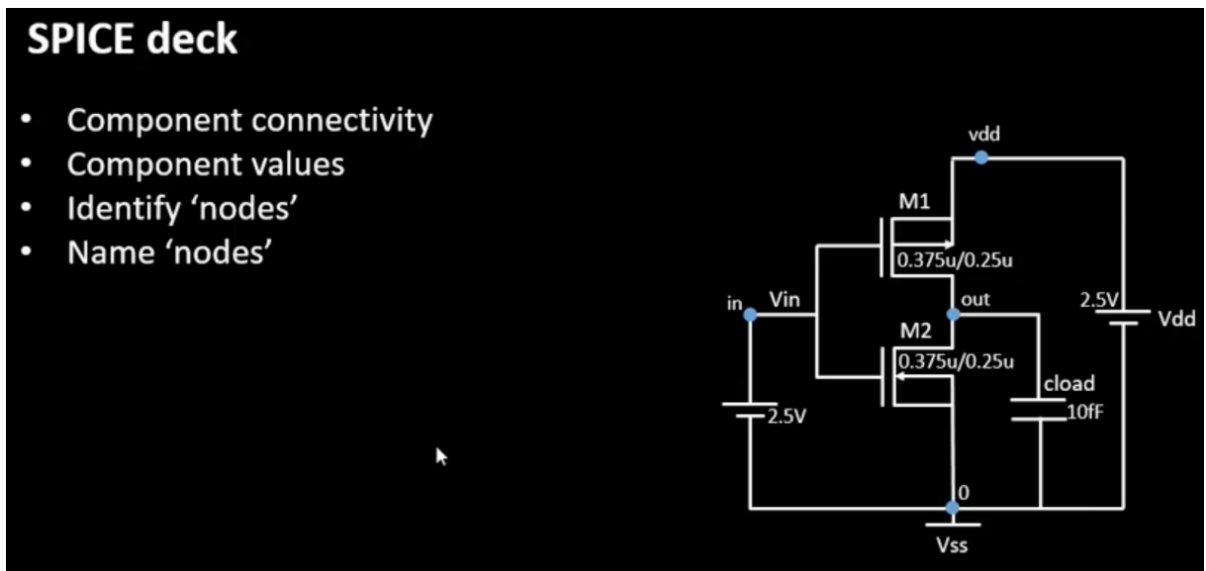


1. Voltage Transfer Characteristics: SPICE Simulations

For a **CMOS inverter**, the **SPICE deck** includes:

- **Component Connectivity:** Define connections between components like **PMOS**, **NMOS**, **Vdd**, **Vss**, and **Vin/Vout**.
- **Component Values:** Specify values for components such as **threshold voltages (Vth)**, **transistor sizes**, and **supply voltages**.
- **Identify Nodes:** Identify all electrical nodes like **Vin**, **Vout**, **source**, **drain**, and **bulk** for both transistors.
- **Name Nodes:** Assign names to each node, ensuring clear identification during simulations (e.g., **Vout** for the output node).



SPICE simulation for CMOS inverter

SPICE Netlist for CMOS Inverter A **SPICE netlist** for a CMOS inverter includes transistor models, power supply connections, input voltage source, transistor specifications (PMOS and NMOS), and simulation commands (e.g., .tran for transient analysis).

SPICE deck

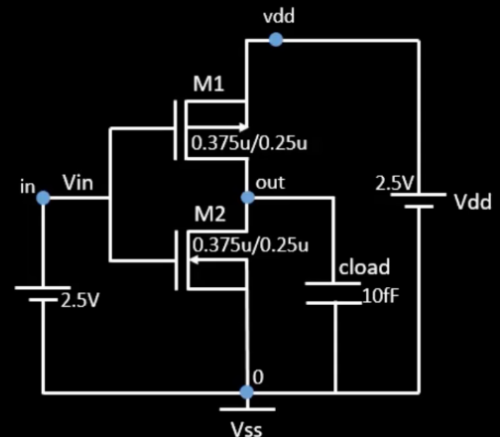
```

*** MODEL Descriptions ***
*** NETLIST Description ***
M1 out in vdd vdd pmos W=0.375u L=0.25u
M2 out in 0 0 nmos W=0.375u L=0.25u

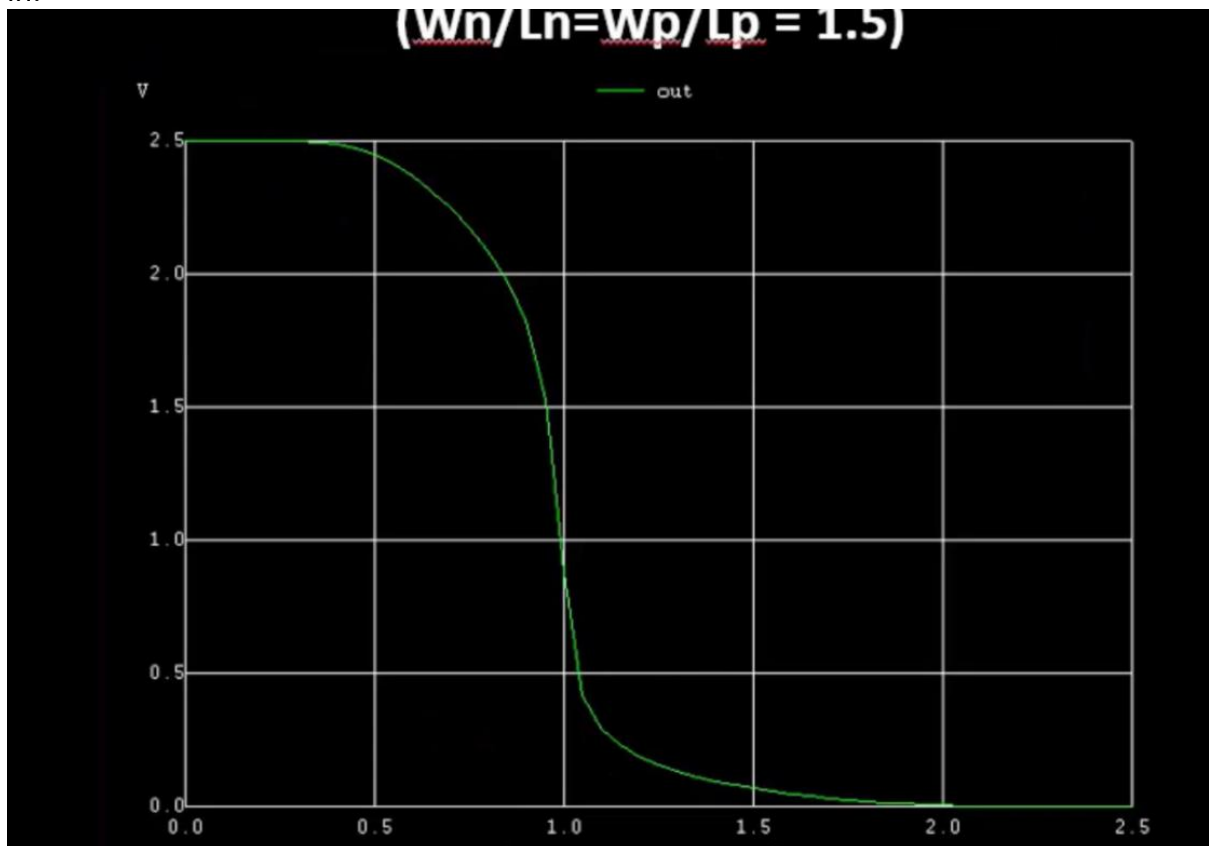
cload out 0 10f

Vdd vdd 0 2.5
Vin in 0 2.5
*** SIMULATION Commands ***
.op
.dc Vin 0 2.5 0.05
*** .include tsmc_025um_model.mod ***
.LIB "tsmc_025um_model.mod" CMOS MODELS
.end

```



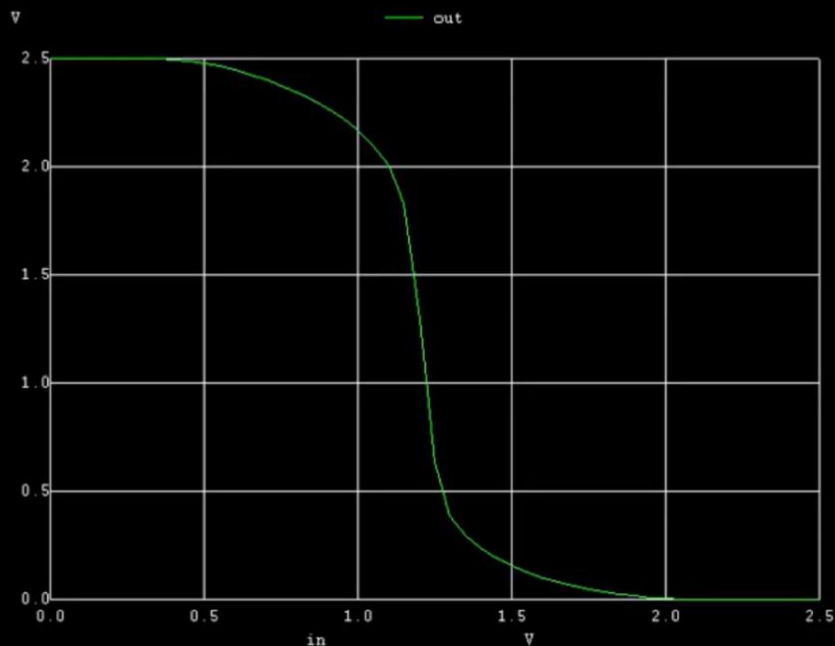
Same $W_n/L_n = W_p/L_p = 1.5$. Plot out vs in:



Now, $W_n/L_n = 1.5$ and $W_p/L_p = 3.75$. Plot out vs

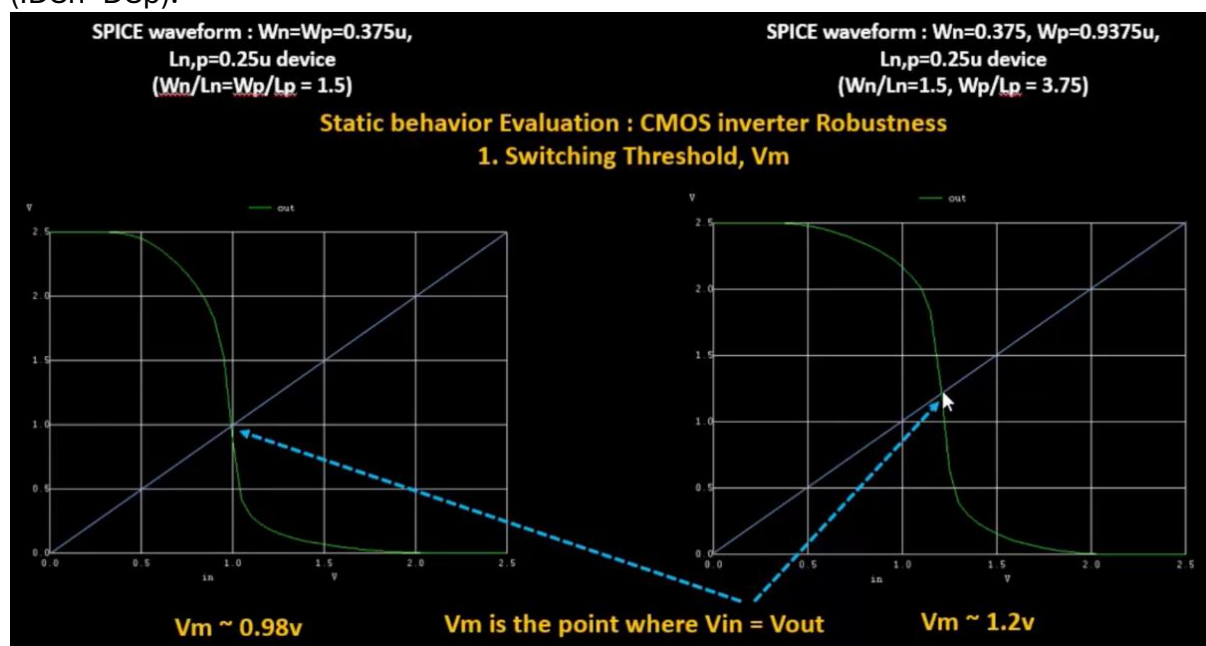
in:

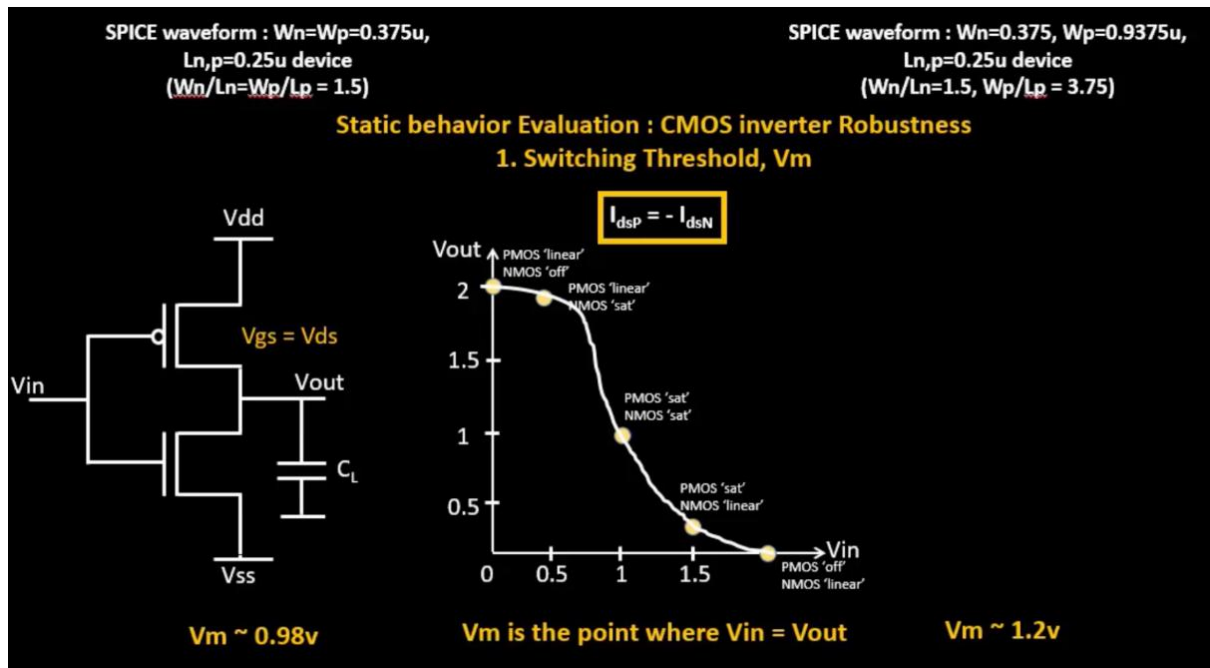
**SPICE waveform : $W_n=0.375$, $W_p=0.9375u$, $L_n,p=0.25u$ device
($W_n/L_n=1.5$, $W_p/L_p = 3.75$)**



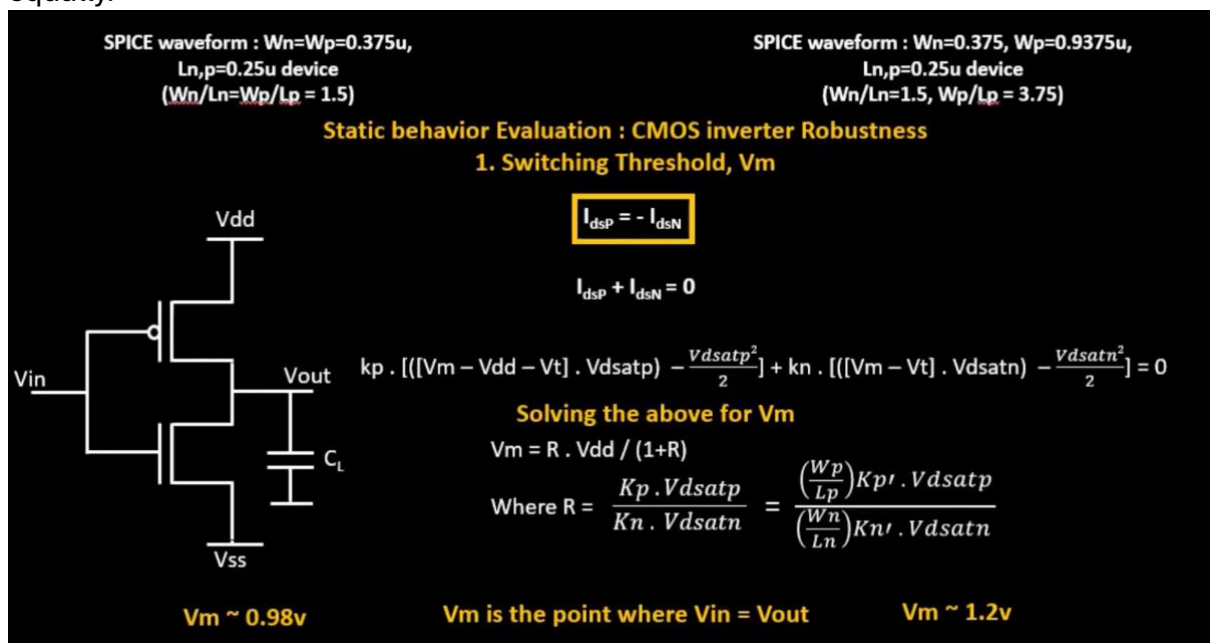
Static Behavior Evaluation: CMOS Inverter Robustness and Switching Threshold (V_m)

The switching threshold (V_m) is the point where $V_{in} = V_{out}$, and both transistors are in saturation (since $V_{ds} = V_{gs}$). At V_m , maximum power is drawn due to large current, and it can be graphically found at the intersection of the VTC with the $V_{in} = V_{out}$ line. The analytical expression for V_m is obtained by equating the drain currents of PMOS and NMOS ($I_{DSn} = I_{DSp}$).





In the **velocity-saturated** case, the **switching threshold (V_m)** is the point where both **NMOS** and **PMOS** transistors are in saturation, and the drain currents are equal. This occurs when the **VDS** of both devices is less than the saturation voltage, i.e., **$V_{DSAT} < (V_m - V_T)$** . The threshold voltage **V_m** can be derived by equating the drain currents of both transistors, with the device widths and lengths (W/L ratios) playing a key role in determining the point where both transistors conduct equally.



Static behavior Evaluation : CMOS inverter Robustness

1. Switching Threshold, V_m

$$I_{dsP} = -I_{dsN}$$

$$I_{dsP} + I_{dsN} = 0$$

$$k_p \cdot \left[([V_m - V_{dd} - V_t] \cdot V_{dsatp}) - \frac{V_{dsatp}^2}{2} \right] + k_n \cdot \left[([V_m - V_t] \cdot V_{dsatn}) - \frac{V_{dsatn}^2}{2} \right] = 0$$

Alternatively, the required ratio of PMOS v/s NMOS transistor size can be derived, such that V_m is set

$$k_n \cdot \left[([V_m - V_t] \cdot V_{dsatn}) - \frac{V_{dsatn}^2}{2} \right] = k_p \cdot \left[([-V_m + V_{dd} + V_t] \cdot V_{dsatp}) + \frac{V_{dsatp}^2}{2} \right]$$

Static behavior Evaluation : CMOS inverter Robustness

1. Switching Threshold, V_m

$$I_{dsP} = -I_{dsN}$$

$$I_{dsP} + I_{dsN} = 0$$

$$k_p \cdot \left[([V_m - V_{dd} - V_t] \cdot V_{dsatp}) - \frac{V_{dsatp}^2}{2} \right] + k_n \cdot \left[([V_m - V_t] \cdot V_{dsatn}) - \frac{V_{dsatn}^2}{2} \right] = 0$$

Alternatively, the required ratio of PMOS v/s NMOS transistor size can be derived, such that V_m is set

$$\frac{K_p \cdot V_{dsatp}}{K_n \cdot V_{dsatn}} = \frac{([V_m - V_t]) - \frac{V_{dsatn}}{2}}{([-V_m + V_{dd} + V_t]) + \frac{V_{dsatp}}{2}}$$

Static behavior Evaluation : CMOS inverter Robustness

1. Switching Threshold, V_m

$$V_m = R \cdot V_{dd} / (1+R)$$

$$\text{Where } R = \frac{K_p \cdot V_{dsatp}}{K_n \cdot V_{dsatn}} = \frac{\left(\frac{W_p}{L_p}\right) K_{p'} \cdot V_{dsatp}}{\left(\frac{W_n}{L_n}\right) K_{n'} \cdot V_{dsatn}}$$

$$\frac{\left(\frac{W_p}{L_p}\right)}{\left(\frac{W_n}{L_n}\right)} = \frac{K_{n'} \cdot V_{dsatn}([V_m - V_t]) - \frac{V_{dsatn}^2}{2}}{K_{p'} \cdot V_{dsatp}([-V_m + V_{dd} + V_t]) + \frac{V_{dsatp}^2}{2}}$$

V_m is the point where $V_{in} = V_{out}$

Velocity Saturation and Switching Threshold (V_m) Analysis

- **Case 1: Velocity Saturation Occurs**
 - Happens in short-channel devices or high supply voltage.
 - Switching threshold **V_m** occurs when currents of PMOS and NMOS transistors are equal.
 - The ratio **r** (PMOS to NMOS strength) influences **V_m** .
 - For **$V_m \approx V_{DD}/2$** , **$r \approx 1$** .
 - **V_m** can be adjusted by changing the PMOS or NMOS width:
 - Increase PMOS width → shift **V_m** upwards.
 - Increase NMOS width → shift **V_m** downwards.
- **Case 2: Velocity Saturation Does Not Occur**
 - Applies to long-channel devices or low supply voltage.
 - The switching threshold **V_m** is still affected by **r** but with a simpler formula.
 - If **$r \approx 1$** , **V_m** is near **$V_{DD}/2$** .
- **PMOS Width Effect on VTC**
 - Increasing PMOS width shifts **V_m** upwards.
 - Increasing NMOS width shifts **V_m** downwards.
 - **V_m** is relatively stable with small variations in transistor ratios.

Static behavior Evaluation : CMOS inverter Robustness

1. Switching Threshold, V_m

W_p/L_p	$x.W_n/L_n$	Rise delay	Fall delay	V_m
W_p/L_p	W_n/L_n	148ps	71ps	0.99v
W_p/L_p	$2W_n/L_n$	80ps	76ps	1.2v
W_p/L_p	$3W_n/L_n$	57ps	80ps	1.25v
W_p/L_p	$4W_n/L_n$	45ps	84ps	1.35v
W_p/L_p	$5W_n/L_n$	37ps	88ps	1.4v

V_m is the point where $V_{in} = V_{out}$

Applications of CMOS inverter in clock network and STA

Static behavior Evaluation : CMOS inverter Robustness

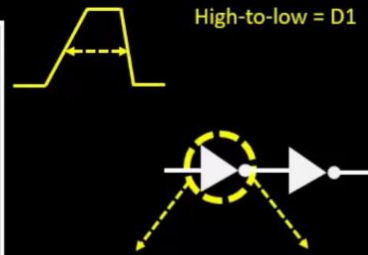
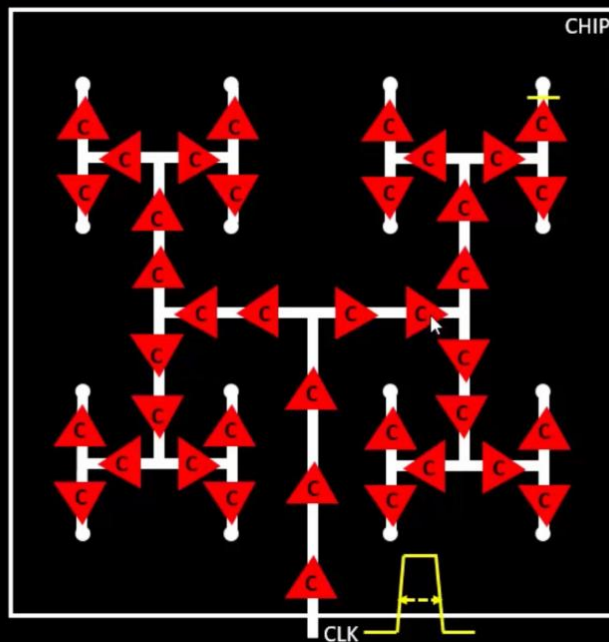
1. Switching Threshold, V_m

W_p/L_p	$x.W_n/L_n$	Rise delay	Fall delay	V_m
W_p/L_p	W_n/L_n	148ps	71ps	0.99v
W_p/L_p	$2W_n/L_n$	80ps	76ps	1.2v
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W_p/L_p	$4W_n/L_n$	45ps	84ps	1.35v
W_p/L_p	$5W_n/L_n$	37ps	88ps	1.4v

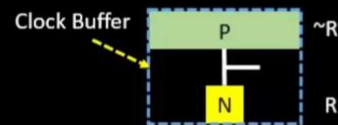
Approximately equal rise-fall delay
Typical characteristics for a clock
inverter/buffer

V_m is the point where $V_{in} = V_{out}$

"Snippet from Clock Tree Synthesis course on Udemy" H-Tree



High-to-low = $D1$



Low-to-high = $D1$

From Device Physics Point of View,
 $R_{ON} \text{ (PMOS)} > R_{ON} \text{ (NMOS)}$ by ~ 2.5 times
 i.e. $R_{ON} \text{ (PMOS)} \sim 2.5 R_{ON} \text{ (NMOS)}$
 For eg. Typical 0.25um technology, and 1V
 If $R_{ON} \text{ (NMOS)} = 40k\Omega$, $R_{ON} \text{ (PMOS)} = 100k\Omega$

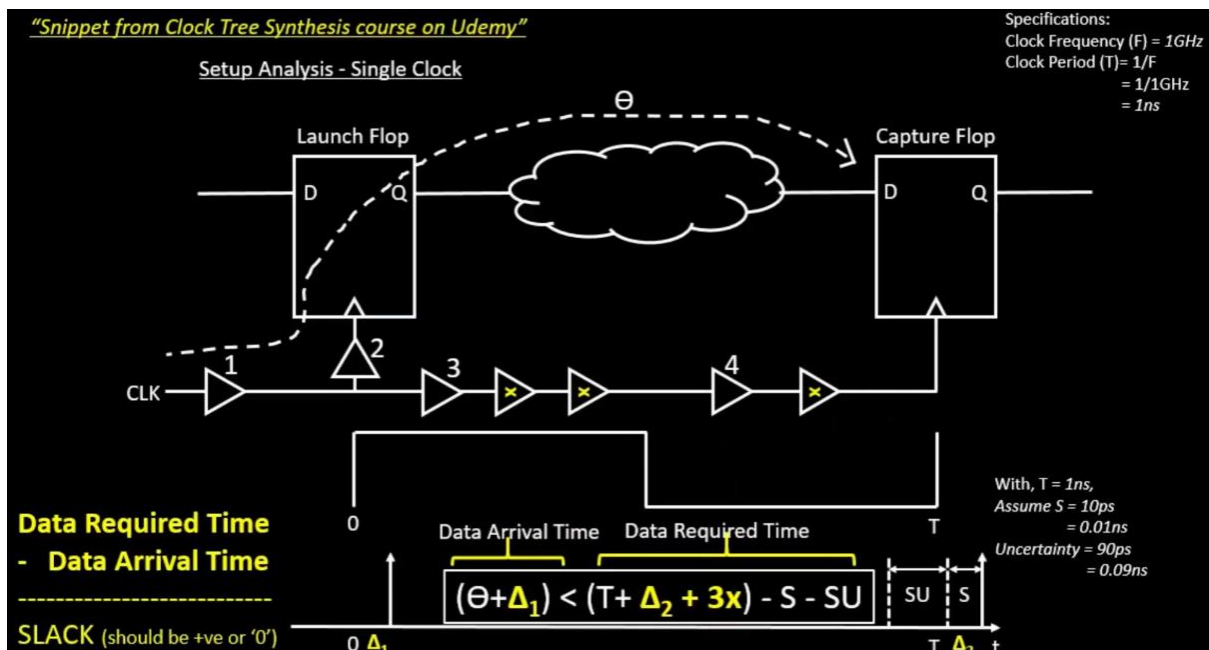
Static behavior Evaluation : CMOS inverter Robustness

1. Switching Threshold, V_m

W_p/L_p	$x \cdot W_n/L_n$	Rise delay	Fall delay	V_m
W_p/L_p	W_n/L_n	148ps	71ps	0.99v
W_p/L_p	$2W_n/L_n$	80ps	76ps	1.2v
W_p/L_p	$3W_n/L_n$	57ps	80ps	1.25v
W_p/L_p	$4W_n/L_n$	45ps	84ps	1.35v
W_p/L_p	$5W_n/L_n$	37ps	88ps	1.4v

Regular inverter/buffer
 Preferred for data path

V_m is the point where $V_{in} = V_{out}$



LABS

1. Voltage transfer characteristics: SPICE simulations

Sky130 SPICE simulation for CMOS - VTC

*Model Description

```
.param temp=27
```

*Including sky130 library files

```
.lib "sky130_fd_pr/models/sky130.lib.spice" tt
```

*Netlist Description

```
XM1 out in vdd vdd sky130_fd_pr__pfet_01v8 w=0.84 l=0.15
```

```
XM2 out in 0 0 sky130_fd_pr__nfet_01v8 w=0.36 l=0.15
```

```
Cload out 0 50fF
```

```
Vdd vdd 0 1.8V
```

```
Vin in 0 1.8V
```

*simulation commands

```
.op
```

```
.dc Vin 0 1.8 0.01
```

```
.control
```

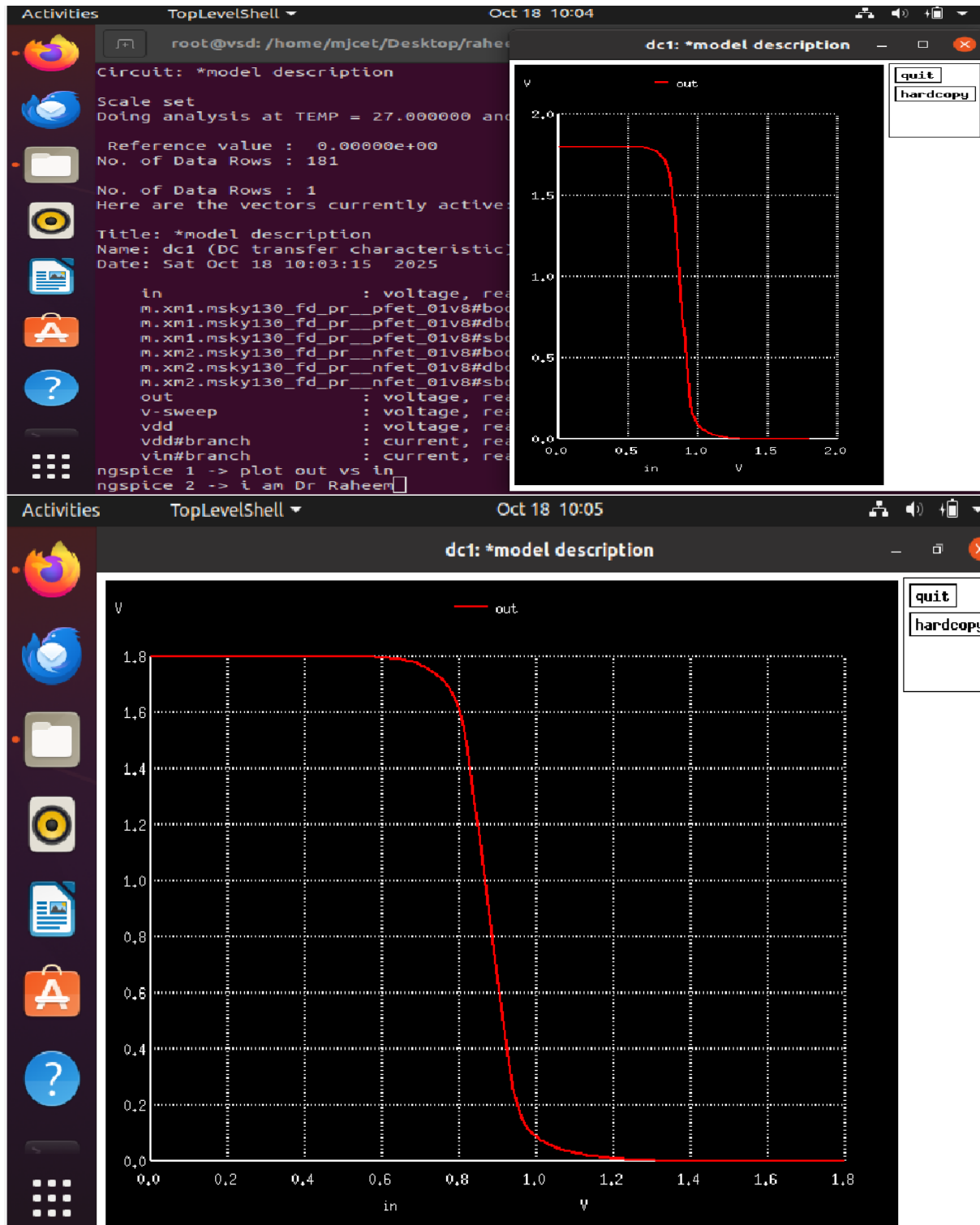
```
run
```

```
setplot dc1
```

```
display
```

```
.endc
```

```
.end
```



Switching Threshold (V_M):

The input voltage (V_{in}) at which the output voltage (V_{out}) equals the input voltage.

- It is the intersection point on the Voltage Transfer Curve (VTC).
- Also referred to as the **midpoint voltage** of the CMOS inverter.

Sky130 SPICE simulation for CMOS - Transient Analysis

*Model Description

```
.param temp=27
```

*Including sky130 library files

```
.lib "sky130_fd_pr/models/sky130.lib.spice" tt
```

*Netlist Description

```
XM1 out in vdd vdd sky130_fd_pr__pfet_01v8 w=0.84 l=0.15
```

```
XM2 out in 0 0 sky130_fd_pr__nfet_01v8 w=0.36 l=0.15
```

```
Cload out 0 50fF
```

```
Vdd vdd 0 1.8V
```

```
Vin in 0 PULSE(0V 1.8V 0 0.1ns 0.1ns 2ns 4ns)
```

*simulation commands

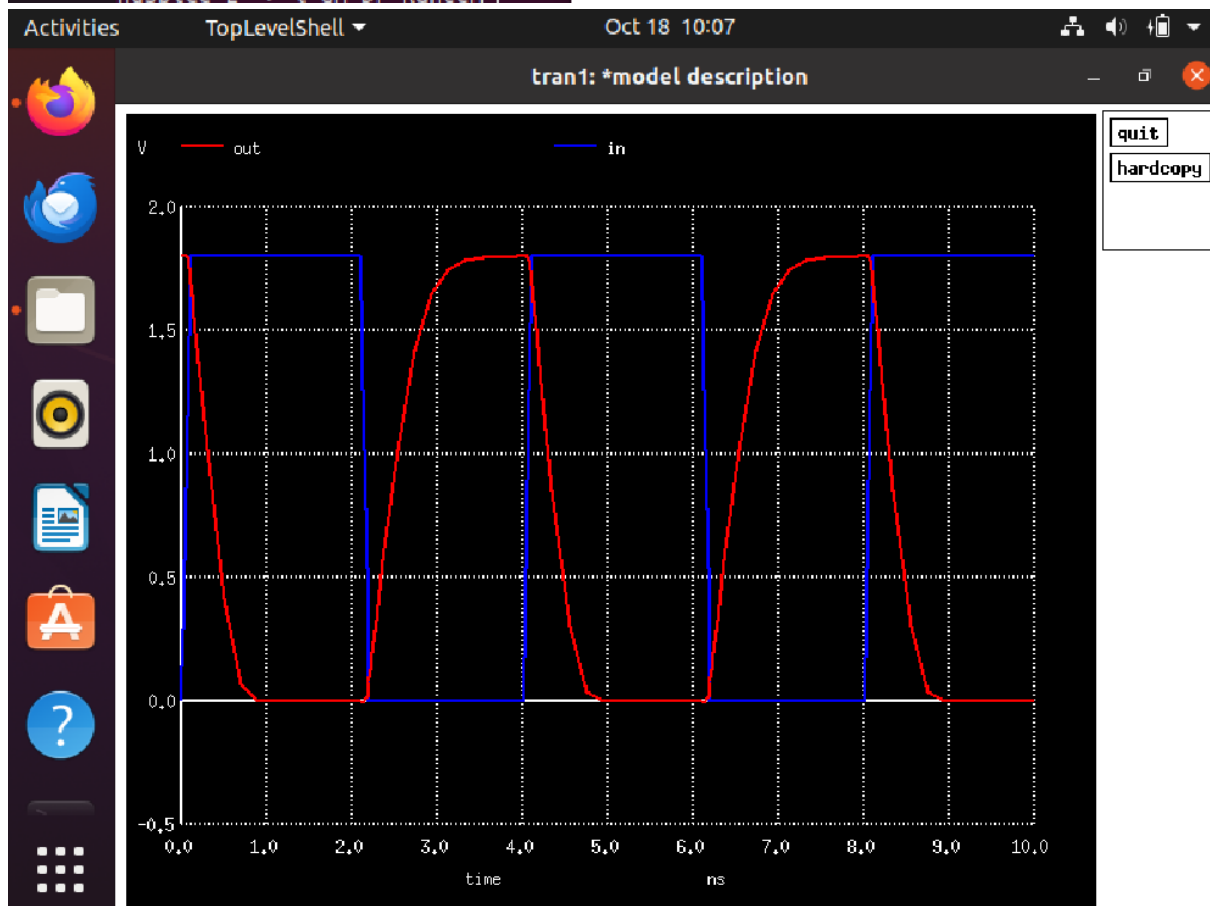
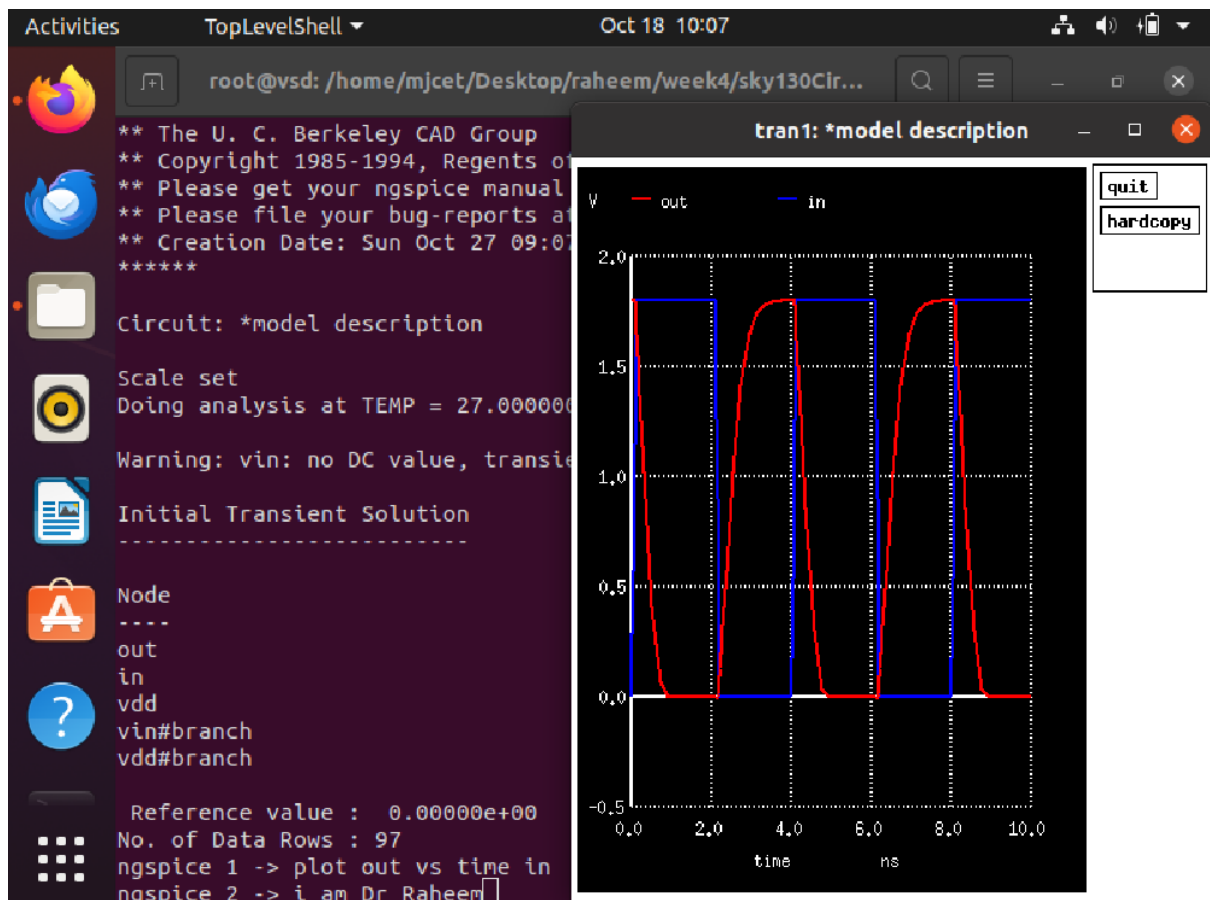
```
.tran 1n 10n
```

```
.control
```

```
run
```

```
.endc
```

.end



Propagation Delay:

The time difference (measured at 50% of input-output transition) between the input signal change and the corresponding output signal switch in a logic gate (e.g., inverter).

Thank you VSD Team for mentoring the RSIC V Reference SoC Tapeout Program

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